Exercise 12.1 SOLUTION

Q1. Find the limit of the following sequences if exists

(i) $a_n = \dfrac{2n+3}{n+1}$

Divide by n:

$$\lim_{n\to\infty} a_n = \lim_{n\to\infty} \frac{\frac{2n}{n}+\frac{3}{n}}{\frac{n}{n}+\frac{1}{n}} = \lim_{n\to\infty} \frac{2+\frac{3}{n}}{1+\frac{1}{n}}$$

As $n\to\infty, \frac{3}{n}\to 0$ *and* $\frac{1}{n}\to 0$:

$$= \frac{2+0}{1+0} = 2$$

(ii) $b_n = \dfrac{2n+3}{n^2+1}$

Divide by n^2:

$$\lim_{n\to\infty} b_n = \lim_{n\to\infty} \frac{\frac{2n}{n^2}+\frac{3}{n^2}}{\frac{n^2}{n^2}+\frac{1}{n^2}} = \lim_{n\to\infty} \frac{\frac{2}{n}+\frac{3}{n^2}}{1+\frac{1}{n^2}}$$

As $n\to\infty, \frac{2}{n}\to 0, \frac{3}{n^2}\to 0$ *and* $\frac{1}{n^2}\to 0$:

$$= \frac{0+0}{1+0} = 0$$

(iii) $c_n = \dfrac{5n^2}{2n+3}$

Divide by n:

$$\lim_{n\to\infty} c_n = \lim_{n\to\infty} \frac{\frac{5n^2}{n}}{\frac{2n}{n}+\frac{3}{n}} = \lim_{n\to\infty} \frac{5n}{2+\frac{3}{n}}$$

As $n\to\infty, \frac{3}{n}\to 0$:

$$= \frac{5(\infty)}{2+0} = \frac{\infty}{2} = \infty$$

The limit does not exist (diverges).

(iv) $d_n = \dfrac{n^2-3n+1}{2n^2+n+4}$

Divide by n^2:

$$\lim_{n\to\infty} d_n = \lim_{n\to\infty} \frac{\frac{n^2}{n^2}-\frac{3n}{n^2}+\frac{1}{n^2}}{\frac{2n^2}{n^2}+\frac{n}{n^2}+\frac{4}{n^2}} = \lim_{n\to\infty} \frac{1-\frac{3}{n}+\frac{1}{n^2}}{2+\frac{1}{n}+\frac{4}{n^2}}$$

As $n\to\infty, \frac{3}{n}\to 0, \frac{1}{n^2}\to 0, and \ \frac{4}{n^2}\to 0$:

$$= \frac{1-0+0}{2+0+0} = \frac{1}{2}$$

Q2. Evaluate each limit by using theorems of limits:

(i) $\lim\limits_{x\to 3}(2x+4)$

Sol: $\lim\limits_{x\to 3}(2x+4) = \lim\limits_{x\to 3}(2x)+\lim\limits_{x\to 3}(4) = 2\lim\limits_{x\to 3}(x)+4$

$= 2(3)+4 = 6+4 = 10$

(ii) $\lim\limits_{x\to 1}(3x^2-2x+4)$

Sol: $\lim\limits_{x\to 1}(3x^2-2x+4) = 3\lim\limits_{x\to 1}(x^2)-2\lim\limits_{x\to 1}(x)+4$

$= 3(1)^2-2(1)+4 = 3-2+4 = 5$

(iii) $\lim\limits_{x\to 3}\sqrt{x^2+x+4}$

Sol: $\lim\limits_{x\to 3}\sqrt{x^2+x+4} = \sqrt{\lim\limits_{x\to 3}x^2+\lim\limits_{x\to 3}x+\lim\limits_{x\to 3}4}$

$= \sqrt{(3)^2+3+4} = \sqrt{9+3+4} = \sqrt{16} = 4$

(iv) $\lim\limits_{x\to 2}\sqrt{x^2+4}$

Sol: $\lim\limits_{x\to 2}\sqrt{x^2+4} = \sqrt{\lim\limits_{x\to 2}x^2+\lim\limits_{x\to 2}4}$

$= \sqrt{(2)^2+4} = \sqrt{4+4} = \sqrt{8}$

(v) $\lim\limits_{x\to 2}\left(\sqrt{x^3+1}-\sqrt{x^2+5}\right)$

Sol: $\lim\limits_{x\to 2}\left(\sqrt{x^3+1}-\sqrt{x^2+5}\right) = \lim\limits_{x\to 2}\sqrt{x^3+1}-\lim\limits_{x\to 2}\sqrt{x^2+5}$

$= \sqrt{(2)^3+1}-\sqrt{(2)^2+5} = \sqrt{8+1}-\sqrt{4+5}$

$= \sqrt{9}-\sqrt{9} = 3-3 = 0$

(vi) $\lim\limits_{x\to -2}\dfrac{2x^3+5x}{3x-2}$

Sol: $\lim\limits_{x\to -2}\dfrac{2x^3+5x}{3x-2} = \dfrac{\lim\limits_{x\to -2}(2x^3+5x)}{\lim\limits_{x\to -2}(3x-2)}$

$= \dfrac{2\lim\limits_{x\to -2}x^3+5\lim\limits_{x\to -2}x}{3\lim\limits_{x\to -2}(x)-2} = \dfrac{2(-2)^3+5(-2)}{3(-2)-2}$

$= \dfrac{2(-8)+5(-2)}{-6-2} = \dfrac{-16-10}{-6-2} = \dfrac{-26}{-8} = \dfrac{13}{4}$

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Q3. Evaluate each limit by using algebraic techniques

نوٹ:
اگر limit put کرنے سے denominator میں zero بنے تو limit put نہیں کرنی اسے آگے حل کرنا ہے اور اگر zero نہ بنے تو یہیں پر limit put کریں۔

(i) $\lim\limits_{x\to -1}\dfrac{x^3-x}{x+1}$

$$=\lim_{x\to -1}\frac{x(x^2-1)}{x+1}=\lim_{x\to -1}\frac{x(x-1)\cancel{(x+1)}}{\cancel{(x+1)}}$$

$$=\lim_{x\to -1}x(x-1)=-1(-1-1)=-1(-2)=2$$

(ii) $\lim\limits_{x\to 3}\dfrac{x^2-5x+6}{x^2-2x-3}$

$$=\lim_{x\to 3}\frac{(x-2)\cancel{(x-3)}}{\cancel{(x-3)}(x+1)}=\lim_{x\to 3}\left(\frac{x-2}{x+1}\right)=\frac{3-2}{3+1}=\frac{1}{4}$$

(iii) $\lim\limits_{x\to 2}\dfrac{x^3-8}{x^2-5x+6}$

$$=\lim_{x\to 2}\frac{x^3-(2)^3}{x^2-3x-2x+6}=\lim_{x\to 2}\frac{(x-2)(x^2+2x+4)}{x(x-3)-2(x-3)}$$

$$=\lim_{x\to 2}\frac{\cancel{(x-2)}(x^2+2x+4)}{\cancel{(x-2)}(x-3)}=\lim_{x\to 2}\frac{x^2+2x+4}{x-3}$$

$$=\frac{(2)^2+2(2)+4}{2-3}=\frac{4+4+4}{-1}=-12$$

(iv) $\lim\limits_{x\to 1}\dfrac{x^3-3x^2+3x-1}{x^3-x}$

$$=\lim_{x\to 1}\frac{x^3-1-3x^2+3x}{x(x^2-1)}$$

$$=\lim_{x\to 1}\frac{(x-1)(x^2+x+1)-3x(x-1)}{x(x-1)(x+1)}$$

$$=\lim_{x\to 1}\frac{\cancel{(x-1)}(x^2+x+1-3x)}{x\cancel{(x-1)}(x+1)}$$

$$=\lim_{x\to 1}\frac{x^2+1-2x}{x(x+1)}=\frac{1+1-2}{1(1+1)}=\frac{2-2}{2}=\frac{0}{2}=0$$

(v) $\lim\limits_{x\to 2}\dfrac{x^3-6x^2+12x-8}{x^3-4x}$

$$=\lim_{x\to 2}\frac{(x-2)^3}{x(x^2-4)}=\lim_{x\to 2}\frac{(x-2)^3}{x(x-2)(x+2)}$$

$$=\lim_{x\to 2}\frac{(x-2)^2}{x(x+2)}=\frac{(2-2)^2}{2(2+2)}=\frac{(0)^2}{2(4)}=\frac{0}{8}=0$$

(vi) $\lim\limits_{x\to 1}\left(\dfrac{x^4-1}{x^2-3x+2}\right)$

$$=\lim_{x\to 1}\left(\frac{x^4-1}{x^2-3x+2}\right)=\lim_{x\to 1}\frac{(x^2-1)(x^2+1)}{x^2-x-2x+2}$$

$$=\lim_{x\to 1}\frac{\cancel{(x-1)}(x+1)(x^2+1)}{\cancel{(x-1)}(x-2)}=\lim_{x\to 1}\frac{(x+1)(x^2+1)}{(x-2)}$$

$$=\frac{(1+1)\left((1)^2+1\right)}{1-2}=\frac{(2)(2)}{-1}=\frac{4}{-1}=-4$$

(vii) $\lim\limits_{x\to 2}\dfrac{x-2}{\sqrt{x+2}-\sqrt{6-x}}$

$$=\lim_{x\to 2}\frac{x-2}{\sqrt{x+2}-\sqrt{6-x}}$$

$$=\lim_{x\to 2}\frac{x-2}{\sqrt{x+2}-\sqrt{6-x}}\times\frac{\sqrt{x+2}+\sqrt{6-x}}{\sqrt{x+2}+\sqrt{6-x}}$$

$$=\lim_{x\to 2}\frac{(x-2)\left(\sqrt{x+2}+\sqrt{6-x}\right)}{\left(\sqrt{x+2}\right)^2-\left(\sqrt{6-x}\right)^2}$$

$$=\lim_{x\to 2}\frac{(x-2)\left(\sqrt{x+2}+\sqrt{6-x}\right)}{x+2-6+x}$$

$$=\lim_{x\to 2}\frac{(x-2)\left(\sqrt{x+2}+\sqrt{6-x}\right)}{2x-4}$$

$$=\lim_{x\to 2}\frac{(x-2)\left(\sqrt{x+2}+\sqrt{6-x}\right)}{2(x-2)}$$

$$=\lim_{x\to 2}\frac{\cancel{(x-2)}\left(\sqrt{x+2}+\sqrt{6-x}\right)}{2\cancel{(x-2)}}$$

$$=\frac{\left(\sqrt{2+2}+\sqrt{6-2}\right)}{2}=\frac{\left(\sqrt{4}+\sqrt{4}\right)}{2}=\frac{(2+2)}{2}=\frac{4}{2}=2$$

(viii) $\lim\limits_{h\to 0}\dfrac{\sqrt{x+h}-\sqrt{x}}{h}$

$$=\lim_{h\to 0}\frac{\sqrt{x+h}-\sqrt{x}}{h}\times\frac{\sqrt{x+h}+\sqrt{x}}{\sqrt{x+h}+\sqrt{x}}$$

$$=\lim_{h\to 0}\frac{\left(\sqrt{x+h}\right)^2-\left(\sqrt{x}\right)^2}{h\left(\sqrt{x+h}+\sqrt{x}\right)}=\lim_{h\to 0}\frac{\cancel{x}+h-\cancel{x}}{h\left(\sqrt{x+h}+\sqrt{x}\right)}$$

$$=\lim_{h\to 0}\frac{\cancel{h}}{\cancel{h}\left(\sqrt{x+h}+\sqrt{x}\right)}=\frac{1}{\sqrt{x+0}+\sqrt{x}}$$

$$=\frac{1}{\sqrt{x}+\sqrt{x}}=\frac{1}{2\sqrt{x}}$$

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(ix) $\lim_{x\to a}\frac{x^n-a^n}{x^m-a^m}$

(divide Numerator and denominator by (x - a))

$$=\frac{\lim_{x\to a}\frac{x^n-a^n}{x-a}}{\lim_{x\to a}\frac{x^m-a^m}{x-a}}$$

$$=\frac{na^{n-1}}{ma^{m-1}}(Use\ Theorem)$$

$$=\frac{n}{m}a^{n-1-m+1}=\frac{n}{m}a^{n-m}$$

Q4. Evaluate the following limits.

(i) $\lim_{x\to 0}\frac{\sin 5x}{x}$

Sol: $\lim_{x\to 0}\frac{\sin 5x}{x}=\lim_{x\to 0}\frac{\sin 5x}{5x}\times 5$ $('\times'\&'\div'by5)$

$=(1)\times 5=5$

Note:

$x^0 = x\times 1^0 = x\times\frac{\pi}{180}=\frac{x\pi}{180}$

(ii) $\lim_{x\to 0}\frac{\sin x^o}{x}$

Sol: $\lim_{x\to 0}\frac{\sin x^o}{x}=\lim_{x\to 0}\frac{\sin\frac{\pi x}{180}}{x}$

$$=\lim_{x\to 0}\frac{\sin x\times\frac{\pi}{180}}{x\times\frac{\pi}{180}}\times\frac{\pi}{180}\quad\left('\times'\&'\div'by\frac{\pi}{180}\right)$$

$$=\lim_{x\to 0}(1)\times\frac{\pi}{180}=\frac{\pi}{180}$$

(iii) $\lim_{\theta\to 0}\frac{1-cos\theta}{sin\theta}$

$$=\lim_{\theta\to 0}\frac{2\,Sin^2\left(\frac{\theta}{2}\right)}{2\,Sin\left(\frac{\theta}{2}\right)Cos\left(\frac{\theta}{2}\right)}=\frac{\lim_{\theta\to 0}Sin\left(\frac{\theta}{2}\right)}{\lim_{\theta\to 0}Cos\left(\frac{\theta}{2}\right)}$$

$$=\frac{Sin0}{Cos0}=\frac{0}{1}=0$$

نوٹ:

کچھ سوال L. Hopital rule سے حل کیے ہیں اگر $\left(\frac{0}{0}\right)$ فارم بن جائے تو Nomerator کا ایک Derivative لینا ہے۔ اور Denomerator کا ایک Derivative لینا ہے۔ پھر اس میں Put limit کرنی ہے۔

(iv) $\lim_{x\to\frac{\pi}{4}}\frac{\sin x-\cos x}{x-\frac{\pi}{4}}$ (First Method)

Ans: $\lim_{x\to\pi/4}\frac{\sin x-\cos x}{x-\pi/4}$

Put $x-\pi/4=\theta\Rightarrow x=\theta+\pi/4$

When $x\to\pi/4$ then $\theta\to 0$

$$=\lim_{\theta\to 0}\frac{\sin\left(\theta+\pi/4\right)-\cos\left(\theta+\pi/4\right)}{\theta}$$

$$=\lim_{\theta\to 0}\frac{\sin\theta\cos\pi/4+\cos\theta\sin\pi/4-\cos\theta\cos\pi/4+\sin\theta\sin\pi/4}{\theta}$$

$$=\lim_{\theta\to 0}\frac{\sin\theta\left(\frac{1}{\sqrt2}\right)+\cos\theta\left(\frac{1}{\sqrt2}\right)-\cos\theta\left(\frac{1}{\sqrt2}\right)+\sin\theta\left(\frac{1}{\sqrt2}\right)}{\theta}$$

$$=\lim_{\theta\to 0}\left(\frac{1}{\sqrt2}\right)\frac{\sin\theta+\cancel{\cos\theta}-\cancel{\cos\theta}+\sin\theta}{\theta}=\lim_{\theta\to 0}\frac{2}{\sqrt2}\left(\frac{\sin\theta}{\theta}\right)$$

$$=\frac{\sqrt2\times\cancel{\sqrt2}}{\cancel{\sqrt2}}\lim_{\theta\to 0}\left(\frac{\sin\theta}{\theta}\right)=\sqrt2(1)=\sqrt2$$

(iv) $\lim_{x\to\frac{\pi}{4}}\frac{\sin x-\cos x}{x-\frac{\pi}{4}}$ (Second Method)

By L. Hopital rule

$$\lim_{x\to\frac{\pi}{4}}\frac{\sin x-\cos x}{x-\frac{\pi}{4}}=\lim_{x\to\frac{\pi}{4}}\frac{\cos x+\sin x}{1-0}$$

$$=\cos\left(\frac{\pi}{4}\right)+\sin\left(\frac{\pi}{4}\right)=\frac{1}{\sqrt2}+\frac{1}{\sqrt2}$$

$$=\frac{2}{\sqrt2}=\frac{\sqrt2.\sqrt2}{\sqrt2}=\frac{\sqrt2.\cancel{\sqrt2}}{\cancel{\sqrt2}}=\sqrt2$$

(v) $\lim_{x\to 0}\frac{\cos ax-\cos bx}{x^2}$ (First Method)

Use formula $\cos p-\cos q=-2\sin\frac{p+q}{2}\sin\frac{p-q}{2}$

$$=\lim_{x\to 0}\frac{-2\sin\frac{ax+bx}{2}.\sin\frac{ax-bx}{2}}{x^2}$$

$$=\lim_{x\to 0}\frac{-2\sin\left(\frac{a+b}{2}\right)x}{x}\cdot\frac{\sin\left(\frac{a-b}{2}\right)x}{x}$$

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$$= -2\lim_{x\to 0}\frac{\sin\left(\frac{a+b}{2}\right)x}{\left(\frac{a+b}{2}\right)x}\left(\frac{a+b}{2}\right).\lim_{x\to 0}\frac{\sin\left(\frac{a-b}{2}\right)x}{\left(\frac{a-b}{2}\right)x}.\left(\frac{a-b}{2}\right)$$

$$= -2(1)\left(\frac{a+b}{2}\right).(1).\left(\frac{a-b}{2}\right) = \cancel{-2}\left(\frac{a^2-b^2}{\cancel{2}\times 2}\right) = \frac{b^2-a^2}{2}$$

(v) $\lim_{x\to 0}\dfrac{\cos ax-\cos bx}{x^2}$ **(Second Method)**

$\lim_{x\to 0}\dfrac{\cos ax-\cos bx}{x^2}$ $\left(\dfrac{0}{0}\right)$

By L. Hopital rule

$$= \lim_{x\to 0}\frac{-a\sin ax+b\sin bx}{2x}$$

$$= \lim_{x\to 0}\frac{-a^2\cos ax+b^2\cos bx}{2}$$

$$= \frac{-a^2\cos 0+b^2\cos(b.0)}{2} = \frac{-a^2+b^2}{2} = \frac{b^2-a^2}{2}$$

(vi) $\lim_{x\to\frac{\pi}{4}}\dfrac{\tan x-1}{x-\frac{\pi}{4}}$ **(First Method)**

Ans: $\lim_{x\to\pi/4}\dfrac{\tan x-1}{x-\pi/4}$

Put $x-\pi/4=\theta \Rightarrow x=\theta+\pi/4$

where $x\to\pi/4$ then $\theta\to 0$

$$= \lim_{\theta\to 0}\frac{\tan\left(\theta+\pi/4\right)-1}{\theta} = \lim_{\theta\to 0}\left(\frac{\tan\theta+\tan\pi/4}{1-\tan\theta\tan\pi/4}\right)-1$$

$$= \lim_{\theta\to 0}\frac{\frac{\tan\theta+1}{1-\tan\theta}-1}{\theta} = \lim_{\theta\to 0}\frac{\tan\theta\cancel{+1}\cancel{-1}+\tan\theta}{1-\tan\theta.\theta}$$

$$= \lim_{\theta\to 0}\frac{2\tan\theta}{\theta}\times\frac{1}{1-\tan\theta} = \lim_{\theta\to 0}2\frac{\sin\theta}{\theta}\frac{1}{\cos\theta}\times\frac{1}{1-\tan\theta}$$

$$= 2\lim_{\theta\to 0}\frac{\sin\theta}{\theta}\times\lim_{\theta\to 0}\frac{1}{\cos\theta}\times\lim_{\theta\to 0}\frac{1}{1-\tan\theta}$$

$$= 2(1)\times\frac{1}{1}\times\frac{1}{1-0} = 2$$

(vi) $\lim_{x\to\frac{\pi}{4}}\dfrac{\tan x-1}{x-\frac{\pi}{4}}$ **(Second Method)**

$\lim_{x\to\frac{\pi}{4}}\dfrac{\tan x-1}{x-\frac{\pi}{4}}$ $\left(\dfrac{0}{0}\right)$

$$= \lim_{x\to\frac{\pi}{4}}\frac{\sec^2 x}{1}$$ By L. Hopital rule

$$= \lim_{x\to\frac{\pi}{4}}\left(\sec^2 x\right)$$

$$= \lim_{x\to\frac{\pi}{4}}\frac{1}{\cos^2 x} = \frac{1}{\left(\cos\left(\frac{\pi}{4}\right)\right)^2} = \frac{1}{\left(\frac{1}{\sqrt{2}}\right)^2} = \frac{1}{\frac{1}{2}} = 2$$

(vii) $\lim_{x\to 0}\dfrac{1-Cos\,2x}{x^2}$

$$= \lim_{x\to 0}\frac{2Sin^2x}{x^2} = 2\left(\lim_{x\to 0}\frac{Sinx}{x}\right)^2 = 2(1)^2 = 2$$

(viii) $\lim_{x\to 0}\dfrac{\cos ax-\cos bx}{\cos cx-\cos dx}$ **(First Method)**

Ans: $\lim_{x\to 0}\dfrac{\cos ax-coxbx}{\cos cx-coxdx}$

$$= \lim_{\theta\to 0}\frac{\cancel{2}\sin\left(\frac{ax+bx}{2}\right)\sin\left(\frac{ax-bx}{2}\right)}{\cancel{2}\sin\left(\frac{cx+dx}{2}\right)\sin\left(\frac{cx-dx}{2}\right)}$$

$$= \frac{\lim\sin\frac{\left(\frac{a+b}{2}\right)x}{\left(\frac{a+b}{2}\right)x}\left(\frac{a+b}{2}\right)\cancel{x}\times\lim\sin\frac{\left(\frac{a-b}{2}\right)x}{\left(\frac{a-b}{2}\right)x}\left(\frac{a-b}{2}\right)\cancel{x}}{\lim\sin\frac{\left(\frac{c+d}{2}\right)x}{\left(\frac{c+d}{2}\right)x}\left(\frac{c+d}{2}\right)\cancel{x}\times\lim\sin\frac{\left(\frac{c-d}{2}\right)x}{\left(\frac{c-d}{2}\right)x}\left(\frac{c-d}{2}\right)\cancel{x}}$$

$$= \frac{(1)\left(\frac{a+b}{2}\right)\times(1)\left(\frac{a-b}{2}\right)}{(1)\left(\frac{c+d}{2}\right)\times(1)\left(\frac{c-d}{2}\right)} = \frac{a^2-b^2}{\cancel{4}}\times\frac{\cancel{4}}{c^2-d^2} = \frac{a^2-b^2}{c^2-d^2}$$

(viii) $\lim_{x\to 0}\dfrac{\cos ax-\cos bx}{\cos cx-\cos dx}$ **(Second Method)**

$\lim_{x\to 0}\dfrac{\cos ax-\cos bx}{\cos cx-\cos dx}$ $\left(\dfrac{0}{0}\right)$

$$= \lim_{x\to 0}\frac{-a\sin ax+b\sin bx}{-c\sin cx+d\sin dx}$$ By L. Hopital rule

$$= \lim_{x\to 0}\frac{-a^2\cos ax+b^2\cos bx}{-c^2\cos cx+d^2\cos dx} = \frac{-a^2\cos(0)+b^2\cos(0)}{-c^2\cos(0)+d^2\cos(0)}$$

$$= \frac{-a^2(1)+b^2(1)}{-c^2(1)+d^2(1)} = \frac{b^2-a^2}{d^2-c^2} = \frac{a^2-b^2}{c^2-d^2}$$

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(ix) $\lim_{x\to 1}\dfrac{x^3-1}{x^2-1}$

$=\lim_{x\to 1}\dfrac{(x-1)(x^2+x+1)}{(x-1)(x+1)}=\dfrac{(1)^2+(1)+1}{1+1}=\dfrac{3}{2}$

(x) $\lim_{x\to 3}\dfrac{x^2-x\log x+3\log x-9}{x-3}$ **(First Method)**

Ans: $\lim_{x\to 3}\dfrac{x^2-x\log x+3\log x-9}{x-3}$

$=\lim_{x\to 3}\dfrac{x^2-9-x\log x+3\log x}{x-3}$

$=\lim_{x\to 3}\dfrac{(x^2-3^2)-\log x(x-3)}{x-3}$

$=\lim_{x\to 3}\dfrac{(x-3)(x+3)-\log x(x-3)}{x-3}$

$=\lim_{x\to 3}\dfrac{(x-3)[x+3-\log x]}{(x-3)}=3+3-\log 3$

$=6-\log 3$

(x) $\lim_{x\to 3}\dfrac{x^2-x\log x+3\log x-9}{x-3}$ **(Second Method)**

Ans: $\lim_{x\to 3}\dfrac{x^2-x\log x+3\log x-9}{x-3}\quad\left(\dfrac{0}{0}\right)$

By L. Hopital rule

$=\lim_{x\to 3}\dfrac{2x-\left(x\left(\frac{1}{x}\right)+1.\log x\right)+\frac{3}{x}-0}{1}$

$=\lim_{x\to 3}\left(2x-1-\ln x+\dfrac{3}{x}\right)=2(3)-\ln(3)-1+\dfrac{3}{3}$

$=6-\ln 3$

(xi) $\lim_{x\to 0}\dfrac{x\left(2^x-1\right)}{1-\cos x}$ **(First Method)**

Ans: $\lim_{x\to 0}\dfrac{x\left(2^x-1\right)}{1-\cos x}$

$=\lim_{x\to 0}\dfrac{x\left(2^x-1\right)}{2\sin^2 x/2}\qquad\left(1-\cos x=2\sin^2 x/2\right)$

$=\lim_{x\to 0}\dfrac{x^2\left(2^x-1\right)}{2x\left(\frac{\sin x/2}{x/2}\cdot\frac{x}{2}\right)}$

$=\dfrac{1}{2}\lim_{x\to 0}\left(x\times\dfrac{2^x-1}{x}\right)\times\dfrac{1}{\lim_{x\to 0}\left(\frac{\sin x/2}{x/2}\right)\times\frac{x}{4}}$

$=\dfrac{1}{2}\lim_{x\to 0}\left(\dfrac{2^x-1}{x}\right)\times\dfrac{4}{1}$

$=2\log e^2=2\log 2\qquad\left(\lim_{x\to 0}\dfrac{2^x-1}{x}=\log e^2\right)$

(xi) $\lim_{x\to 0}\dfrac{x\left(2^x-1\right)}{1-\cos x}$ **(Second Method)**

$\lim_{x\to 0}\dfrac{x\left(2^x-1\right)}{1-\cos x}\quad\left(\dfrac{0}{0}\right)$

By L. Hopital rule

$=\lim_{x\to 0}\dfrac{\left(2^x-1\right)+x2^x\ln 2}{\sin x}\qquad\left(\dfrac{0}{0}\right)$

$=\lim_{x\to 0}\dfrac{2^x\ln 2-0+2^x\ln 2+x\ln 2.2^x\ln 2}{\cos x}$

$=\lim_{x\to 0}\dfrac{2.2^x\ln 2+x.2^x(\ln 2)^2}{\cos x}$

$=\dfrac{2.2^0\ln 2+0.2^0(\ln 2)^2}{\cos(0)}$

$=\dfrac{2\ln 2}{1}=2\ln 2$

Q5. Express each limit in terms of e:

(i) $\lim_{n\to+\infty}\left(1+\dfrac{1}{n}\right)^{2n}$

$=\left[\lim_{n\to+\infty}\left(1+\dfrac{1}{n}\right)^n\right]^2=e^2$

(ii) $\lim_{n\to+\infty}\left(1+\dfrac{1}{n}\right)^{n/2}$

$=\left[\lim_{n\to+\infty}\left(1+\dfrac{1}{n}\right)^n\right]^{\frac{1}{2}}=e^{\frac{1}{2}}=\sqrt{e}$

(iii) $\lim_{n\to+\infty}\left(1-\dfrac{1}{n}\right)^n=\lim_{n\to+\infty}\left[1+\left(-\dfrac{1}{n}\right)\right]^n$

$=\left[\lim_{n\to+\infty}\left(1+\left(\dfrac{-1}{n}\right)\right)^{-n}\right]^{-1}=e^{-1}=\dfrac{1}{e}$

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(iv) $\lim_{n\to+\infty}\left(1+\frac{1}{3n}\right)^{n}$

$$=\left[\lim_{n\to+\infty}\left(1+\frac{1}{3n}\right)^{3n}\right]^{\frac{1}{3}}=e^{\frac{1}{3}}$$

(v) $\lim_{n\to+\infty}\left(1+\frac{4}{n}\right)^{n}$

$$=\lim_{n\to+\infty}\left(1+\frac{4}{n}\right)^{4n/4}=\left[\lim_{n\to+\infty}\left(1+\frac{4}{n}\right)^{n/4}\right]^{4}=e^{4}$$

(vi) $\lim_{x\to0}(1+3x)^{2/x}$

$$=\lim_{x\to0}(1+3x)^{\frac{6}{3x}}=\left[\lim_{x\to0}(1+3x)^{1/3x}\right]^{6}=e^{6}$$

(vii) $\lim_{x\to0}(1+2x^2)^{\frac{1}{x^2}}$

$$=\lim_{x\to0}\left(1+2x^2\right)^{\frac{2}{2x^2}}=\left[\lim_{x\to0}\left(1+2x^2\right)^{1/2x^2}\right]^{2}=e^{2}$$

(viii) $\lim_{x\to0}\frac{e^{ax}-e^{bx}}{abx}$ **(First Method)**

Ans: $\lim_{x\to0}\frac{e^{ax}-e^{bx}}{abx}$

$$=\lim_{x\to0}\frac{e^{ax}-e^{bx}-1+1}{abx}$$

$$=\lim_{x\to0}\frac{e^{ax}-1}{abx}-\frac{e^{bx}-1}{abx}$$

$$=\frac{1}{b}\lim_{x\to0}\frac{e^{ax}-1}{ax}-\frac{1}{a}\lim_{x\to0}\frac{e^{bx}-1}{bx}$$

$$=\frac{1}{b}e^{e}-\frac{1}{a}e^{e}$$

$$=\frac{1}{b}\ln e-\frac{1}{a}\ln e=\frac{1}{b}(1)-\frac{1}{a}(1)$$

$$=\frac{1}{b}-\frac{1}{a}$$

(viii) $\lim_{x\to0}\frac{e^{ax}-e^{bx}}{abx}$ **(Second Method)**

By L. Hopital rule

$$=\lim_{x\to0}\frac{ae^{ax}-be^{bx}}{ab}=\frac{ae^{0}-be^{0}}{ab}=\frac{a-b}{ab}=\frac{1}{b}-\frac{1}{a}$$

(ix) $\lim_{x\to\infty}\left(\frac{x}{1+x}\right)^{x}=\lim_{x\to\infty}\left(\frac{1+x}{x}\right)^{-x}$

$$=\lim_{x\to\infty}\left(\frac{1}{x}+\frac{x}{x}\right)^{-x}$$

$$=\lim_{x\to\infty}\left(\frac{1}{x}+1\right)^{-x}=\left[\lim_{x\to\infty}\left(1+\frac{1}{x}\right)^{x}\right]^{-1}=e^{-1}=\frac{1}{e}$$

(x) $\lim_{x\to0}\frac{e^{1/x}-1}{e^{1/x}+1}, x<0,$

$\Rightarrow$ *Put* $x=-y$ *when* $x\to0$ *then* $y\to0$

$$=\lim_{y\to0}\frac{e^{1/-y}-1}{e^{1/-y}+1}=\lim_{y\to0}\frac{\frac{1}{e^{1/y}}-1}{\frac{1}{e^{1/y}}+1}=\frac{\frac{1}{\infty}-1}{\frac{1}{\infty}+1}=\frac{0-1}{0+1}=-1$$

(xi) $\lim_{x\to0}\frac{e^{1/x}-1}{e^{1/x}+1}, x>0$

$$=\lim_{x\to0}\frac{e^{1/x}\left(1-\frac{1}{e^{1/x}}\right)}{e^{1/x}\left(1+\frac{1}{e^{1/x}}\right)}=\frac{1-\frac{1}{\infty}}{1+\frac{1}{\infty}}=\frac{1-0}{1+0}=1$$

(xii) $\lim_{x\to2}\frac{e^{x}-e^{2}}{x-2}$ $\left(\frac{0}{0}\right)$ **(First Method)**

Ans: $\lim_{x\to2}\frac{e^{x}-e^{2}}{x-2}$

Put $x-2=y\Rightarrow x=y+2$

When $x\to2$ then $y\to0$

$$=\lim_{y\to0}\frac{e^{y+2}-e^{2}}{y}=\lim_{y\to0}\frac{e^{y}.e^{2}-e^{2}}{y}$$

$$=\lim_{y\to0}\frac{e^{2}\left(e^{y}-1\right)}{y}=e^{2}\lim_{y\to0}\frac{e^{y}-1}{y}$$

$$=e^{2}\log e^{e}=e^{2}\ln e=e^{2}(1)=e^{2}$$

(xii) $\lim_{x\to2}\frac{e^{x}-e^{2}}{x-2}$ $\left(\frac{0}{0}\right)$ **(Second Method)**

Ans: By L. Hopital rule

$$=\lim_{x\to2}\frac{e^{x}}{1}=e^{2}$$

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